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# Aqua Well Robot Solution (AWRS)

Single descent diagnostics: 360° imaging, in site water quality, casing integrity, gas / liquid leaks detection, multi depth water sampling

IP Application No. SA 1020252223

Supported by Energy & Water Academy and Tuwaiq Academy

## Problem Statement:

- Groundwater wells are often monitored with outdated rope-and-bailer tools, giving limited insight into groundwater condition and wellbore integrity.
- These methods miss depth-dependent water-quality changes and multi-depth aquifer chemistry.
- They can overlook early casing corrosion, structural damage, and toxic gas/chemical leaks.
- Worldwide, many wells are already contaminated or unreliable, increasing the need for better monitoring.
- Without real-time, multi-depth monitoring, issues may go unnoticed until failures, risking water safety and raising costs.

Figure 1: Traditional methods require attendance of work team

Figure 2: Traditional rope-and-bailer groundwater sampling method.

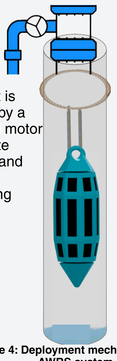
## Solution:

AWRS is an integrated smart well-inspection platform that combines multiple structural, chemical, and environmental sensors into one automated device deployed in a single descent. It delivers complete real-time diagnostics in one operation.

In addition, a novel AI-powered EIS model was developed to detect and classify minerals directly inside the well.

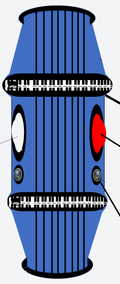
Figure 3: AWRS autonomous well-inspection robot.

## How it works?



the robot is lowered by a rope and motor for remote descent and distant measuring

Figure 4: Deployment mechanism of the AWRS system.



Laser beam sensor to check thickness of well pipe

Rotating section to allow rotation of the middle part of the AWRS robot

XRF sensor shoots a laser beam against metal to check its integrity and elements.

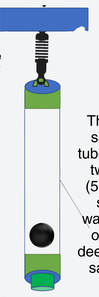
High definition camera allows live feed of the pipe condition

Figure 5: Internal sensor layout and rotating inspection module.

The robot's lower analytical section uses dual hydraulic arms to simultaneously collect a 500 ml surface sample and a 1 L deep-water sample. Once filled, the arms retract the tubes into the chassis for operator retrieval.

gas sensor (H2S, CO2, H2 and CH4 and SO2)

Figure 6: Gas sensing and analytical module.



This is the sampling tube that has two sizes (500 ml for surface water and 1 or 2 L for deeper water sampling.

Figure 7: Dual-volume water sampling tube.

## Minerals detection

**Target Selection**
Lead (Pb) selected as primary target due to its high toxicity. Future expansion will include more minerals.

**EIS Measurement**
Impedance Analyzer performs EIS on a modified gold sensor. Six replicate readings per concentration across 100 Hz 200 mHz.

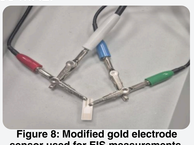


Figure 8: Modified gold electrode sensor used for EIS measurements.

**Data Processing & Feature Extraction**
Gamry Echem Analyst processes raw data. Equivalent Circuit Modeling extracts Rct, Rs, and CPE as ML feature vectors.

Figure 9: Laboratory setup for electrochemical data acquisition and feature extraction.

**ML Model Evaluation**
Seven classifiers benchmarked: KNN, Decision Tree, GBC, AdaBoost, Logistic Regression, SVC, and Random Forest.

**Model Selection**
Random Forest Classifier (RFC) selected as final model for superior accuracy across all tested concentrations.

## Results

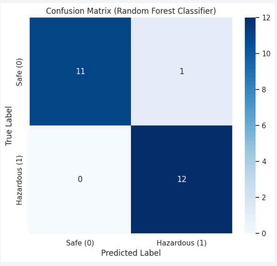
Approach Validation: The results confirmed the success of using Machine Learning to decode the Electrochemical Impedance Spectroscopy (EIS) data. The highest accuracy achieved was 96% (using the RFC model), which fulfils high-accuracy spectrum classification.
Model Selection: Linear (LR) and basic (SVC) models failed to achieve the required accuracy (54% - 71%). In contrast, Ensemble Methods, especially Random Forest, demonstrated a superior capability to handle the nature of EIS data, indicating that the relationship between the extracted features and the concentration is non-linear and complex.

| ML Model | Classification Accuracy |
|---|---|
| Random Forest Classifier (RFC) | 0.96 (96%) |
| Gradient Boosting Classifier (GBC) | 0.92 (92%) |
| AdaBoost Classifier (ABC) | 0.92 (92%) |
| Decision Tree (DT) | 0.88 (88%) |
| Logistic Regression (LR) | 0.71 (71%) |
| K-Nearest Neighbors (KNN) | 0.71 (71%) |
| Support Vector Classifier (SVC) | 0.54 (54%) |

**Table 1:**

| Data # | Sampled date | Sample ID | Aquifer Name | Temperature ($^{o}$C) | pH | Pressure Psi | $O_2$ % | $CO_2$ PPM | $H_2S$ PPM | Humidity % | TDS (mg/L) | Water Level distance (ft) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | September 12, 2025 | 00123 | Arab-C | 20 | 7.1 | 10 | 90 | 400 | 0 | 10 | 1000 | 45 |
| 2 | September 12, 2025 | 001234 | Arab-C | 40 | 7.2 | 15 | 90 | 700 | 0 | 45 | 2000 | 50 |
| 3 | September 13, 2025 | 0012345 | Arab-C | 60 | 7.25 | 23 | 90 | 924 | 0 | 80 | 2500 | 110 |

**Table 2:**




Performance Analysis: The final performance of the adopted Random Forest Classifier was rigorously assessed using a Confusion Matrix. As shown in Figure 4, this matrix is critical for analyzing the model's reliability in critical detection scenarios. The low rate of False Negatives (FN) where the system fails to flag a truly hazardous sample confirms the high degree of safety and reliability, ensuring the system meets high-standard requirements for heavy metal detection.

## Operational Comparison

| Traditional methods | Our method |
|---|---|
| Team of 3–5 technicians | Remotely controlled from distance |
| Random spot rope-and-bailer sampling | monitors water quality in site and also take high accurate multi depth sealed samples |
| Separate gas detection tool | Detect gasses + liquids in one run |
| Lab transport for mineral testing | Detect minerals in the water directly |
| The process takes days and small problems go unnoticed till it escelates | Process takes minutes |

## Objectives

- Detects early casing damage (corrosion, erosion, cracks) to prevent failures
- Identifies hazardous gas/chemical leaks sooner to protect groundwater
- Improves monitoring accuracy with multi-depth water-quality assessment
- Cuts time-on-site and labor by replacing multiple sequential tool deployments with one integrated run
- Enhances operator safety by minimizing manual/field exposure
- Supports proactive maintenance and faster response to contamination risks
- AI-based mineral identification using EIS

## References:

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